

Daril Vergesse Sobze Soffack

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EDUCATION

Technical University Darmstadt

Darmstadt, Germany

Bachelor of Science in Computer Science

Expected Oct. 2026

- **Concentrations:** AI/Machine Learning, Data Systems
- **Related Coursework:** Functional and Object-Oriented Programming (1.0), Linear Algebra (1.7), Analysis and Combinatorics (1.3), Machine Learning (2.7), Computer Networks and Distributed Systems (1.3)
- **Seminars & Labs:** AI for Data Engineering, Computing for Social Good

EXPERIENCE

IDAT GmbH

Darmstadt, Germany

AI Engineer Intern

March 2025 – Present

- Developed and productionized a Retrieval-Augmented Generation (RAG) System for enterprise knowledge search and Q&A across internal data sources.
- Designed and implemented custom data connectors for SharePoint, ClickUp, and internal websites, indexing 10,000+ documents into a vector database.
- Enabled 30+ employees to retrieve company documents and ask contextual questions in <5 seconds, significantly reducing manual search time.
- Integrated the support tickets database (3000+ tickets), enabling natural-language queries and automated daily ingestion to keep 100% of data updated.
- Delivered technical presentations on AI, ML, and GenAI to 40+ colleagues to improve internal AI adoption.

Technical University Darmstadt

Darmstadt, Germany

Teaching Assistant - Data Structures and Algorithms

April 2026 – Sept. 2026

- Led weekly tutorial sessions for undergraduate students, explaining complex algorithmic concepts, Big O notation, data structure and algorithms implementations.
- Evaluated and graded programming assignments and exams, providing constructive feedback to improve students' coding proficiency.

Fraunhofer SIT

Darmstadt, Germany

Bachelor Student

April 2026 – present

- Researching feature fusion techniques combining Self-Supervised Learning (SSL) and Spectral Features to improve the detection and source tracing of synthetic audio.
- Building a detector that not just verify if an audio is fake or genuine but also determines the source, generation's method and the model

Studienkreis GmbH

Fürth, Germany

Mathematics Tutor

March 2024 – May 2024

- Supported students' academic growth in mathematics by fostering confidence and independence, resulting in more self-reliant learners.

PROJECTS

HYDRA: Data Analysis Automation for Medicinal Molecules | Team Lead

Oct. 2025 – Present

- Automating protein mass spectrometry data analysis using signal processing and ML to accelerate drug discovery.
- Developing a Python application to analyze MS1 data with 708+ subfiles, including Chromatograms extraction and Gaussian fitting.

CALLPILOT AI | Founder (Hack-Nation AI Hackathon, MIT)

Feb. 2026

- Developed an Autonomous Agentic System that scans for local service providers, makes automated phone calls, and schedules appointments directly in Google Calendar.

HR Orchestrator Agentic AI | Founder (IBM Dev Day, Germany)

Jan. 2026 – Feb. 2026

- Built an autonomous hiring system using "IBM WATSONX ORCHESTRATE" with three specialized mini-agents.
- Automated job posting to Slack, resume scoring against requirements, and automated email communication (invitations/rejections) for candidates.

PAPERS

AI By Hand: Information Retrieval

Sept. 2025

- Computed TF-IDF vector spaces and Cosine Similarity matrices from scratch to evaluate the mathematical foundations of natural language search algorithms.

SELECT THEN MATCH? A new LLM-powered Entity Matching

Feb. 2025

- Optimized an NLP entity resolution framework by identifying and mitigating "false positive bias" in LLM list-wise selection prompts.
- Architected a novel "Select-Then-Match" pipeline that decouples candidate ranking from 6-shot verification, boosting the overall F1-score from 86.56 to 95.49 using GPT-5-Mini.

Extending the CAESURA Framework: Modernization, Robustness, and Fine-Tuning

Feb. 2025

- Engineered a pipeline for a multi-modal query planner that automatically intercepts execution traces to generate ChatML/JSONL datasets for Small Language Model (SLM) fine-tuning
- Modernized the core LLM infrastructure by integrating open-weight models (Llama-3.3-70B), reducing average API latency by 75% and token costs by 60%.

TECHNICAL SKILLS

Programming: Java, Python, HTML/CSS, SQL, C++

Tools: Figma, Bash, Scikit-learn, PyTorch, Docker, REST APIs, RAG, OpenAI API, Vector Databases, Tkinter, Flask

Languages: English (Business Fluent), German (Business Fluent), French (Native)

ACTIVITIES AND LEADERSHIP

SV Kamerun Darmstadt 1999

Vice-Captain

Darmstadt, Germany

July 2024 – Present